# Noah Mok

163 University Ave W, Waterloo, ON, N2L 3E5
780-937-8298 | noahtmok@gmail.com | linkedin.com/in/ntmok | noahmok.github.io

## Education

**Bachelor of Applied Science, Mechanical Engineering** | University of Waterloo | 2026 - 2031

## Technical Skills

**CAD**: **SolidWorks**, **Siemens NX**, **AutoCAD**
**Coding Languages**: **Python** (Sympy, Matplotlib, Numpy, Tkinter), **C++**, HTML, CSS, JavaScript
**Technical Knowledge**: **Dimensioning and Tolerancing, Solid Modeling, Surface Modeling**, Data Analysis, Statics and Kinematics
**Machining and Manufacturing**: Drill press, Lathe, Milling machine, Bandsaw, Press Machine
**Microsoft 365**: Teams, Outlook, Word, PowerPoint, Excel (VBA, Power BI)
**Additional**: Alberta Class 5 Drivers License

## Experience

**University of Waterloo Automotive Future Team** | Sep 2026 – active
*University of Waterloo*

- Lead research on traction capabilities and benchmarking for UWAFT's competition vehicle, the 2026 Jeep Cherokee, and for alternative off-roading vehicle competitors to design traction objectives and optimize vehicle performance.
- Designed a rotary selector switch and mount in **Siemens NX** for integration into the 2026 competition vehicle, validating it with static shear and pull-out calculations to meet safety requirements.

**University of Waterloo Midnight Sun** | Sep 2026 – active
*University of Waterloo*

- Created and manufactured front light mounts using 3D print filament to secure the front lights of the vehicle such that they meet competition constraints and can withstand vibration from the car.
- Designed the **Solar Car Canopy** in **SolidWorks** to keep the driver and car components safe, optimize aerodynamic performance, and practice **surface modeling** tools in 3D CAD.

**Applied Mathematics Researcher** | Jun 2022 & Jun 2019
*MathPath & Epsilon Math Camp*

- **Led a team of 16 people** in researching isoperimetric sets of English words with Prof. Chip Morgan using **data analysis and Python** to reverse engineer the process behind online tools like auto-correct, published in *Parabola*, Vol. 59, Issue 2.
- Co-developed a solution on Watt's linkage motion with Dr. Zoltan Kovacs, published in the *International Journal of Technology in Mathematics Education*, Vol. 27, No. 4.

**Student Page** | Sep 2023 – Oct 2025
*Edmonton Public Library | Edmonton, AB*

- Worked 10-12 hours a week with a team of student pages demonstrating a service-oriented mindset, organizing, shelving, and sorting book returns and arrivals and assisting customers in navigating software such as **Word, PowerPoint, and Outlook**.

## Projects

**Solar Car Canopy** | Oct 2026

- Designed a solar car canopy using **Surface Modeling** in **SolidWorks** as part of an onboarding challenge for the University of Waterloo Midnight Sun team.
- Crafted an aerodynamic canopy to meet dimensioning constraints, and researched and performed **data analysis** on previous aerobody designs to reduce estimated drag area by **60 percent** from the MS 15 canopy.

**Rotary Selector Switch and Mount** | Sep 2026

- Designed a rotary selector switch and its mount using **Siemens NX** as part of an onboarding challenge for the University of Waterloo Automotive Future Team.
- Created orthographic views in **AutoCAD** and a design document detailing my solutions to the **dimensioning, safety, and manufacturing** constraints for integration into a production vehicle.
- Improved safety, accessibility, and lifetime of the switch while reducing estimated material costs by **20 percent** compared to the previous iteration of the mount in the 2022 competition vehicle.

**Custom Computer Mouse** | Oct 2026

- Researched into current market trends with computer mice to create a mouse which fit practical constraints and had a consumer-friendly design.
- Utilized **surface and solid modeling tools** in **SolidWorks** to create a computer mouse with a mouse wheel and side buttons to meet my idealized gaming mouse design.

**Kinematics Calculator** | Jul 2026

- Developed a desktop GUI application through **Python** to solve 1D and 2D kinematic equations to help students with complex physics calculations.
- Utilized **SymPy** to dynamically substitute known values and solve reduced equation systems, saving upwards of **15 minutes per calculation**.